

AI-BFSI Internship Project Documentation

Solution Architecture – Subscription Waste Detector

Executive Summary

Solution Architecture defines the high-level structure of a software system and illustrates how different components work together to achieve the project objectives.

The Subscription Waste Detector follows a modular architecture consisting of:

- **Streamlit-based user interface**
- **Python processing modules**
- **Pandas-based data analysis**
- **Subscription detection logic**
- **Plotly visualizations**
- **Groq API integration** for AI-generated recommendations

Each component performs a specific responsibility, resulting in a simple, maintainable, and efficient application.

Introduction

The architecture of a software system serves as the blueprint for development. It identifies the major components, defines their responsibilities, and illustrates the flow of information between them.

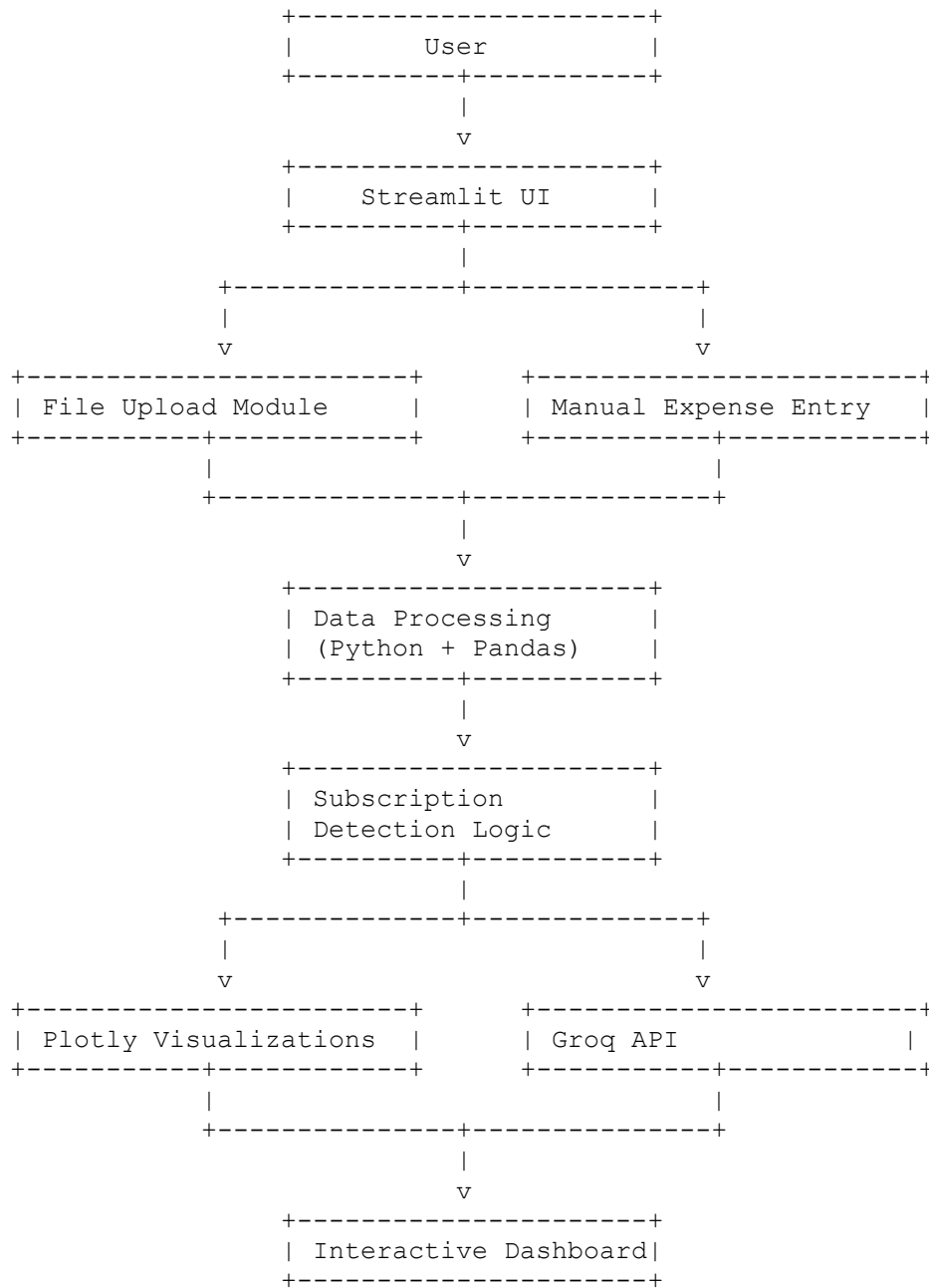
The Subscription Waste Detector adopts a lightweight architecture suitable for a Streamlit application. Instead of relying on complex distributed systems, the application processes uploaded expense data locally within the Streamlit environment before generating visualizations and AI-powered recommendations.

This architecture was selected because it satisfies the functional requirements while remaining simple to develop, test, and maintain.

Architecture Goals

- Maintain a modular software structure
- Process uploaded expense records efficiently
- Detect recurring subscription payments accurately
- Present financial insights through interactive dashboards
- Generate AI-powered recommendations using the Groq API
- Support future enhancements without requiring major architectural changes

High-Level Architecture



Architecture Components

1. User Interface Layer

Technology: Streamlit Responsibilities:

- Upload CSV/Excel files
- Manual expense entry
- Display dashboard components
- Present AI recommendations

2. Data Processing Layer

Technology: Python + Pandas Responsibilities:

- Read uploaded files
- Clean input data
- Organize transaction records
- Prepare data for subscription analysis

3. Subscription Detection Layer

Responsibilities:

- Detect recurring transactions
- Group subscription expenses
- Calculate recurring costs
- Prepare subscription summaries

4. Visualization Layer

Technology: Plotly Responsibilities:

- Monthly spending charts
- Subscription summaries
- Expense visualizations
- Dashboard analytics

5. AI Recommendation Layer

Technology: Groq API (Llama model via Groq) Responsibilities:

- Generate AI-powered recommendations
- Suggest subscription optimization
- Improve financial awareness

Note: The application does **not** train its own AI model. It uses Groq API for inference only.

Data Flow

1. User uploads expense data or enters transactions manually.
2. Streamlit receives the input.
3. Python processes the uploaded records.
4. Pandas organizes and analyzes the transaction data.
5. Subscription detection logic identifies recurring subscriptions.
6. Plotly generates dashboard visualizations.
7. Subscription summaries are sent to the Groq API.
8. AI recommendations are returned.
9. Results are displayed through the Streamlit dashboard.

Technology Architecture

Layer	Technology
User Interface	Streamlit
Programming Language	Python
Data Processing	Pandas
Spreadsheet Support	OpenPyXL
Visualization	Plotly
AI Integration	Groq API
Development Environment	VS Code

Architecture Characteristics

- **Simplicity:** Modular design reduces complexity.
- **Maintainability:** Each component has a single responsibility.
- **Scalability:** Future modules can be added without redesign.
- **Reusability:** Processing modules can be reused in other projects.
- **Modularity:** Clear separation of layers improves organization.

Architecture Limitations

Version 1 intentionally excludes:

- Database layer
- User authentication
- Cloud deployment
- REST API backend
- Multi-user session management
- Background task scheduling

Future Architecture Enhancements

Potential expansions include:

- SQLite or PostgreSQL database
- User authentication
- Cloud storage
- REST API services
- Email notification service
- Multi-user support
- Bank API integration

Conclusion

The Solution Architecture of the Subscription Waste Detector provides a modular and practical foundation for analyzing recurring subscription expenses. By combining Streamlit, Python, Pandas, Plotly, and Groq API, the system delivers an efficient workflow that transforms uploaded expense data into meaningful financial insights.

The architecture remains intentionally lightweight, focusing on Version 1 functionality while supporting future expansion. This ensures maintainability, scalability, and alignment with the